Marcel David Baroi

Full Stack Engineer · TypeScript / Vue 3 / NestJS · PHP / Laravel · PostgreSQL / Supabase

Dhaka, Bangladesh (UTC+6) · Remote preferred · Open to on-site/hybrid in Dhaka

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SUMMARY

Product-minded full stack engineer with nearly 2 years of professional experience and a 3.91/4.00 CS degree, with a track record of independently architecting and shipping complex, data-intensive business applications. Fluent in TypeScript (Vue 3, NestJS, Next.js) with strong relational database fundamentals (PostgreSQL, RLS, atomic transactions). Engineered and maintain a multi-tenant wholesale ERP in daily production (6,000+ SKUs) and the Service Desk platform embedded across company client apps.

- **Engineered** a multi-tenant ERP in daily production (6,000+ SKUs); client roughly doubled order volume after adoption.
- **Sole maintainer** of a company-wide Service Desk platform, embedded as the standard support plugin across all client apps.
- **Replaced** paper warehouse audits with a barcode-scanning Capacitor Android app synced via atomic database RPCs.

EXPERIENCE

Software Engineer (Full Stack) — Nov 2024 – Present

Proficient Information Systems · Dhaka, Bangladesh · On-site / hybrid

Product engineering team of 40+ developers and 10+ admin/staff delivering multi-tenant SaaS and client software. Scope below reflects solo ownership of two production systems.

- **Service Desk (Support Platform) — Sole Engineer:** Designed and shipped the company's support platform (Vue 3/Quasar, Laravel, MongoDB, MySQL) — standalone web portal plus an embeddable package now the standard support plugin across all company client applications.
- **Unified Intake & Deduplication:** Consolidated three support channels into one inbox — in-app Vue widget, IMAP email sync (using subject and sender matching for threading), and public web forms hardened with double opt-in, rate limiting (Laravel: 3/min, 100/hr), and reCAPTCHA.
- **Automated Issue Context:** Client widget captures current route state, recent API calls, and client-side runtime errors on ticket submit; server-side sanitization strips sensitive payloads before persistence, making tickets instantly reproducible.
- **Package Infrastructure:** Co-built the company-wide private npm packaging workflow (Vite library mode, SemVer). Collaborated with a teammate to prototype approaches, presented to senior engineers, and integrated the approved solution (auth, ticketing, common components) to cut setup time on new client apps.
- **SuperCards — Task Management Module:** Contributed to the task-management feature within a 10+ engineer team (mostly mid/senior); refactored core components into private packages for cross-project reuse.
- **Mobile (Solo):** Built the Flutter frontend for a microfinance loan-officer app (daily disbursements/collections) and a Quasar-based loan-collection app shipped as a Capacitor Android build.

Contract Full Stack Engineer — Multi-Tenant Wholesale ERP — 2025 – Present

(part-time contract, alongside full-time role)

Engaged by an import-distribution business to design and build a custom production ERP. I designed the system; the business uses it in daily production.

- **Business Impact:** Moved order-taking, billing, and inventory tracking fully out of spreadsheets into an automated system; the client roughly doubled order volume and increased sales after adoption.
- **Complexity & Operations:** Manages 3 tenants, a live catalog of 6,000+ products, and regular monthly processing of 20+ wholesale and 60+ dropship orders. ~3 admin users across 3 businesses, each serving ~15–20 B2B customers. Bills and invoices generate automatically on order confirmation.
- **Origin & Ownership:** While working part-time at the business, identified their manual Excel-based workflow and proposed, designed, and solo-built the custom ERP (Vue 3, Quasar, TypeScript, Supabase, PostgreSQL).
- **Strict Tenant Isolation:** Enforced tenant isolation and access controls in PostgreSQL using Row-Level Security (RLS) policies and PL/pgSQL security functions keyed to `auth.uid()` and membership checks — providing defense-in-depth beyond application-level filtering.
- **Inventory & Transactional Integrity:** Mitigated overselling across B2B storefront, dropshipping, and direct wholesale channels using atomic Supabase RPCs, row-level locks (`SELECT FOR UPDATE`), and Available-To-Promise (ATP) logic.

- **Billing & Financial Ledger:** Engineered a billing and cashbook system supporting standard bills, debit/credit notes, and payment allocations. Built a landed-cost calculation engine that snapshots FX rates and allocates freight/customs duties based on product weight at shipment finalization.
- **Warehouse Mobile App:** Replaced paper warehouse audits with a Capacitor Android app: continuous barcode scanning (Google ML Kit), discrepancies synced to PostgreSQL via atomic RPCs.
- **Frontend Performance:** Implemented route-level code splitting and lazy loading for heavy UI tables and charting libraries; utilized Pinia and TanStack Query for aggressive caching and cursor pagination to keep interactions responsive on low-bandwidth networks.
- **Observability & CI/CD:** Established GitHub Actions workflows for automated testing and Vercel auto-deploy; implemented Sentry for frontend/backend error tracking with email alerting. Configured Supabase daily backups with documented restore procedures.
- **Security & Dependency Management:** Managed secrets via environment variables and server-only service-role keys; tracked dependency CVEs via manual pnpm audits.

PROJECTS

MindVault · Double-Entry Finance API & Client 2025

Backend · Frontend — Handwritten to extend backend stack beyond Supabase/Laravel

- Architected a double-entry ledger API with a decimal-precise immutable transaction engine, per-account role-based access control (RBAC), and automated integration tests using **NestJS, TypeORM, PostgreSQL**.
- Built a responsive web client with **Next.js 15, TypeScript, Tailwind CSS** to visualize net worth and cash-flow analytics.

TECHNICAL SKILLS

Frontend	Vue 3, Quasar, TypeScript, JavaScript, Pinia, TanStack Query, Tailwind CSS, Next.js, HTML5/CSS3
Mobile	Flutter (frontend), Capacitor, Android builds, Google ML Kit
Backend & APIs	TypeScript, Node.js, NestJS, PHP, Laravel, REST API design
Databases	PostgreSQL / Supabase (schema design, RLS, PL/pgSQL RPCs), MySQL, MongoDB, TypeORM
Testing & CI/CD	PHPUnit/Laravel test cases, Jest/integration tests, GitHub Actions, Vercel, Sentry
Tooling	Git, Docker, Vite, pnpm/npm private registries, modular/private packages, Linux, Cloudflare Pages

EDUCATION

B.Sc. in Computer Science & Engineering 2020 – 2024

Daffodil International University · CGPA: 3.91 / 4.00 · HackerRank Verified: Python, JavaScript, React, SQL

Higher Secondary Certificate (HSC), Science 2017 – 2019

Notre Dame College, Dhaka · GPA: 4.67 / 5.00